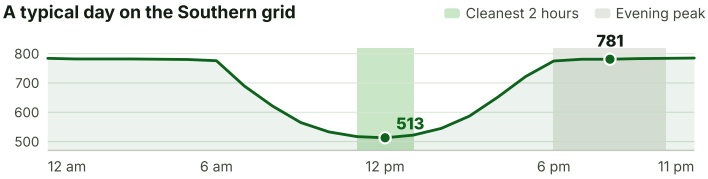
SRM INSTITUTE OF SCIENCE & TECHNOLOGY
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ZeroEmit: AI-Powered Carbon-Aware EV Charging

Charge your EV when India's grid is cleaner.

Problem statement

- **An EV is only as clean as the power that charges it,** and most of India's electricity still comes from coal.
- **Grid carbon intensity swings through the day:** on the Southern grid, 513 gCO_2/kWh at noon but 781 at 8 pm.
- **Drivers can't see this,** so they charge by habit, often in the evening, when each kWh emits the most CO_2.



gCO_2/kWh by hour (IST), averaged over the 14 days to 1 Nov 2026. Source: India Energy Atlas.

Proposed solution

ZeroEmit is a free web app that EV drivers install on their phone. It reads the live carbon intensity of the driver's regional grid, forecasts the next 24 hours and tells them when to plug in.

Know the best time
The cleanest 2-hour windows for today and tomorrow, for the driver's city.

Get reminded
A notification before the best window starts.

See the impact
Each logged charge shows the CO_2 it avoided and earns points for good timing.

Key features

Live grid intensity
All 5 regional grids, every state and 255 cities.

24-hour forecast
Hourly curve and the 3 cleanest windows each day.

Smart reminders
Best-window alerts, a daily digest and streak nudges.

Charge log
kWh, grid intensity, cost and CO_2 avoided per charge.

Rewards
Eco points, 4 levels, 8 badges and a monthly leaderboard.

Charger finder
Public EV chargers nearby on a map, with directions.

Impact card
A one-tap image of the month's CO_2 avoided.

Private and offline
History stays on the phone; works without signal.

Built with

Next.js 16 · React 19 · TypeScript · PWA · Web Push · Upstash Redis · Vercel · GitHub Actions · Leaflet · GSAP · Lenis

Methodology and system architecture

Data sources
India Energy Atlas grid data
Electricity Maps fallback
OpenStreetMap chargers

→ **Server on Vercel**
Next.js API routes
Upstash Redis cache
Scheduler every 15 min

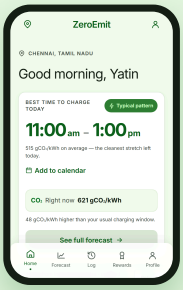
→ **Carbon-aware decision engine**
Forecast, rank, nudge, score
Explainable rules on real grid data

→ **ZeroEmit app**
React 19 web app
Works offline
Push notifications

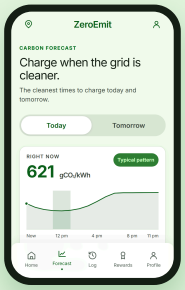
1. **Read** the current carbon intensity of the driver's regional grid, cached on the server.
2. **Forecast** a typical day: the average for each hour over the last 14 days, with the live reading for the current hour.
3. **Rank** every 2-hour window in the next 24 hours by average intensity; keep the 3 cleanest that don't overlap.
4. **Nudge** with a push reminder before the best window, and an alert when now is the cleanest time left.
5. **Score** each logged charge: +10 points in a best window, +5 at a cleaner-than-average hour. CO_2 avoided = (650 − intensity at charging) × kWh.

Dashboard

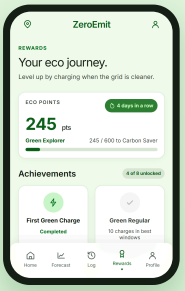
Screens from the live app, shown with sample data.



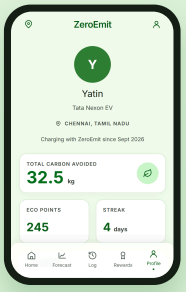
Home
Today's best time to charge.



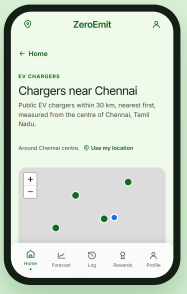
Forecast
The day's curve, best window.



Rewards
Points, levels and badges.



Impact
CO_2 avoided, points and streak.



Chargers
Public EV chargers nearby.

Impact

781 (8 pm) → **513** (12 pm) Moving one 30 kWh charge from 8 pm to noon on the Southern grid avoids about **8 kg of CO_2**.

Illustrative: (781 − 513) gCO_2/kWh × 30 kWh, from the chart's data. Not a measured result.

- **Lower emissions per kWh** from timing alone, on chargers drivers already use.
- **Less charging in the evening peak,** when the grid leans hardest on coal.
- **Visible, lasting habits:** monthly CO_2 avoided, streaks and badges for each driver.

SDG mapping

SDG 7 — Affordable and Clean Energy
Shifts charging to hours with more clean power, such as midday solar.

SDG 11 — Sustainable Cities and Communities
Cleaner urban EV transport, with data for 255 cities.

SDG 13 — Climate Action
Cuts the CO_2 behind every charge and shows drivers what they avoided.

Team

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Scan to open ZeroEmit